01.04 · THEORY + PAST-PAPER INDEX

Spectroscopy and Atomic Physics

Reading the fingerprints carried by starlight

START HERE

Spectroscopy turns light into evidence about atoms, motion, temperature and material between the stars. This lecture builds the atomic picture first, then explains continua and spectral lines, Doppler shifts, line broadening and polarization.

LEARNING GOALS

✓	Explain how quantized atomic energies create absorption and emission lines.
✓	Recognize continuum, emission-line and absorption-line spectra.
✓	Infer line-of-sight velocity from a Doppler shift and state the limits.
✓	Separate line shifts from broadening and describe common broadening mechanisms.
✓	Use polarization as a clue to scattering and source geometry.

1 · Atoms, energy levels and photons

An atom contains a small, positively charged nucleus surrounded by electrons. Quantum mechanics allows an electron in an atom to occupy only certain energy states. The allowed energies depend on the element and its physical environment. A transition between two states must exchange a photon whose energy matches the energy difference.

$$|\Delta E| = h\nu = hc/\lambda$$

When an electron moves to a higher-energy state, the atom absorbs a photon. When it moves to a lower-energy state, the atom emits a photon. Since each element has its own pattern of energy levels, its set of spectral lines acts like a fingerprint. Ionization, pressure and magnetic fields can modify the pattern.

WAVELENGTH AND ENERGY

A larger transition energy corresponds to a higher-frequency, shorter-wavelength photon. Use $E(\text{eV}) \approx 1240/\lambda(\text{nm})$ as a convenient conversion. The same transition has the same rest wavelength in emission and absorption, although its observed wavelength can be shifted by motion or altered by the source environment.

2 · Continuous spectra, emission lines and absorption lines

A hot, dense object produces a broad, nearly continuous spectrum because many interactions blend together. A thin, hot gas can produce bright emission lines at particular wavelengths. If a continuous source is viewed through a cooler, thinner gas, atoms in the gas remove photons at selected wavelengths and create dark absorption lines.

THREE IDEALIZED SPECTRAL SITUATIONS

Dense hot material → continuous spectrum. Thin hot gas against a dark background → emission lines. Continuous light passing through cooler gas → absorption lines. Real stars combine these effects: a photospheric continuum is modified by absorption in their atmospheres, and hot outer gas can add emission.

A spectrum is a plot of intensity against wavelength or frequency. The continuum shape carries information about temperature and extinction; line positions identify elements and reveal motion; line depths and ratios depend on temperature, density, ionization and the number of atoms along the path.

A line is not a perfectly narrow spike. Atoms move, collide and have finite lifetimes, while telescopes and spectrographs have finite resolving power. These effects spread the signal over a small range of wavelength.

3 · Doppler shifts: motion along the line of sight

If a source moves toward or away from an observer, the spacing of its wave crests changes. For speeds much smaller than c, the fractional wavelength shift is approximately the radial velocity divided by c. Define $\Delta\lambda = \lambda_{observed} - \lambda_{rest}$. With this convention positive radial velocity means recession and a redshift; negative means approach and a blueshift.

$$v_r/c \approx \Delta\lambda/\lambda_{rest} \cdot \Delta\nu/\nu_{rest} \approx -v_r/c$$

A single spectral shift measures only the component of velocity along the line of sight. It cannot reveal the transverse motion without other information such as proper motion. For speeds comparable to c, the classical approximation fails and a relativistic Doppler relation is needed.

AVOID SIGN CONFUSION

Always write down the rest wavelength and compute observed minus rest before interpreting the sign. A line observed at a longer wavelength is redshifted and, in the low-speed approximation, the source is receding. A shift of a whole line profile is different from the line merely becoming wider.

4 · Why spectral lines broaden

Thermal Doppler broadening comes from random motions of emitting or absorbing atoms: atoms moving toward us contribute slightly shorter wavelengths and atoms moving away contribute slightly longer wavelengths. Higher gas temperature produces a broader distribution of velocities. Turbulent motions can add another Doppler contribution.

Pressure or collisional broadening occurs when close encounters perturb atomic energy levels and shorten the time an atom emits or absorbs coherently. Natural broadening follows from the finite lifetime of an excited state. Rotation broadens a stellar line because one limb approaches while the other recedes. Instrumental broadening is imposed by the spectrograph and detector.

$$\text{Resolving power } R = \lambda/\Delta\lambda$$

Here $\Delta\lambda$ is the smallest wavelength separation the instrument can distinguish near λ. A larger R means finer spectral resolution. If a line is Doppler-broadened by a characteristic speed v, its fractional width is roughly $\Delta\lambda/\lambda \approx v/c$, but exact definitions depend on the width convention and line profile.

SHIFT VERSUS WIDTH

A common velocity of the whole source shifts the line centre. A spread of velocities broadens the line. A line may be shifted and broadened at the same time; fit or inspect the centre and width separately.

5 · Scattering and polarization

Light is a transverse electromagnetic wave: its electric field oscillates perpendicular to its direction of travel. Unpolarized light has electric-field directions that vary without a preferred orientation. A linear polarizer transmits one orientation and absorbs or rejects the perpendicular component.

$$\text{Malus's law: } I = I_0 \cos^2\theta$$

Malus's law applies when the incident beam is already linearly polarized and θ is the angle between its polarization direction and the analyzer axis. Unpolarized light passing through an ideal first polarizer has average intensity $I_0/2$. Scattering can polarize light because the scattered intensity depends on the direction of the electric-field oscillation relative to the scattering geometry.

ASTRONOMICAL USE

Polarization can reveal scattering by dust, reflection from a surface, magnetic-field geometry or an asymmetric source. It is a vector property of light, so the instrument's orientation and the sky position angle matter. Ordinary intensity alone does not specify polarization.

6 · A practical spectrum-reading workflow

Calibrate wavelength using known reference lines, subtract the detector background, and identify the continuum before measuring lines. Compare candidate rest wavelengths in a consistent unit. For a small shift, calculate $\Delta\lambda/\lambda_rest$, multiply by c, and state whether the result is radial velocity, line width or instrumental resolution.

Use multiple lines when possible: a single feature can be misidentified, blended or affected by calibration error. Compare line ratios only after checking that the lines are not saturated and that dust extinction has been considered. Report uncertainties and significant figures that match the

data.

WHAT A SPECTRUM CANNOT SAY BY ITSELF

A spectrum does not automatically give the full three-dimensional velocity, distance or unique temperature. Those inferences need a model, additional observables or independent data. Distinguish a directly measured quantity from a quantity inferred under assumptions.

PAST IOAA PROBLEM REFERENCES

The following questions are from official past IOAA problem sets.

Use each year, round and question label to locate the original problem in its official paper.

01	IOAA 2007 · TH 1.8
02	IOAA 2007 · TH 1.12
03	IOAA 2008 · TH L2
04	IOAA 2009 · TH P2
05	IOAA 2012 · TH S9
06	IOAA 2013 · TH S13
07	IOAA 2014 · TH P7
08	IOAA 2014 · TH P8
09	IOAA 2015 · TH S6
10	IOAA 2015 · TH S8
11	IOAA 2015 · DA D2
12	IOAA 2016 · TH T8
13	IOAA 2017 · TH T4
14	IOAA 2018 · TH T6
15	IOAA 2019 · TH 5
16	IOAA 2021 · TH Q4
17	IOAA 2023 · TH Q2
18	IOAA 2025 · TH T04